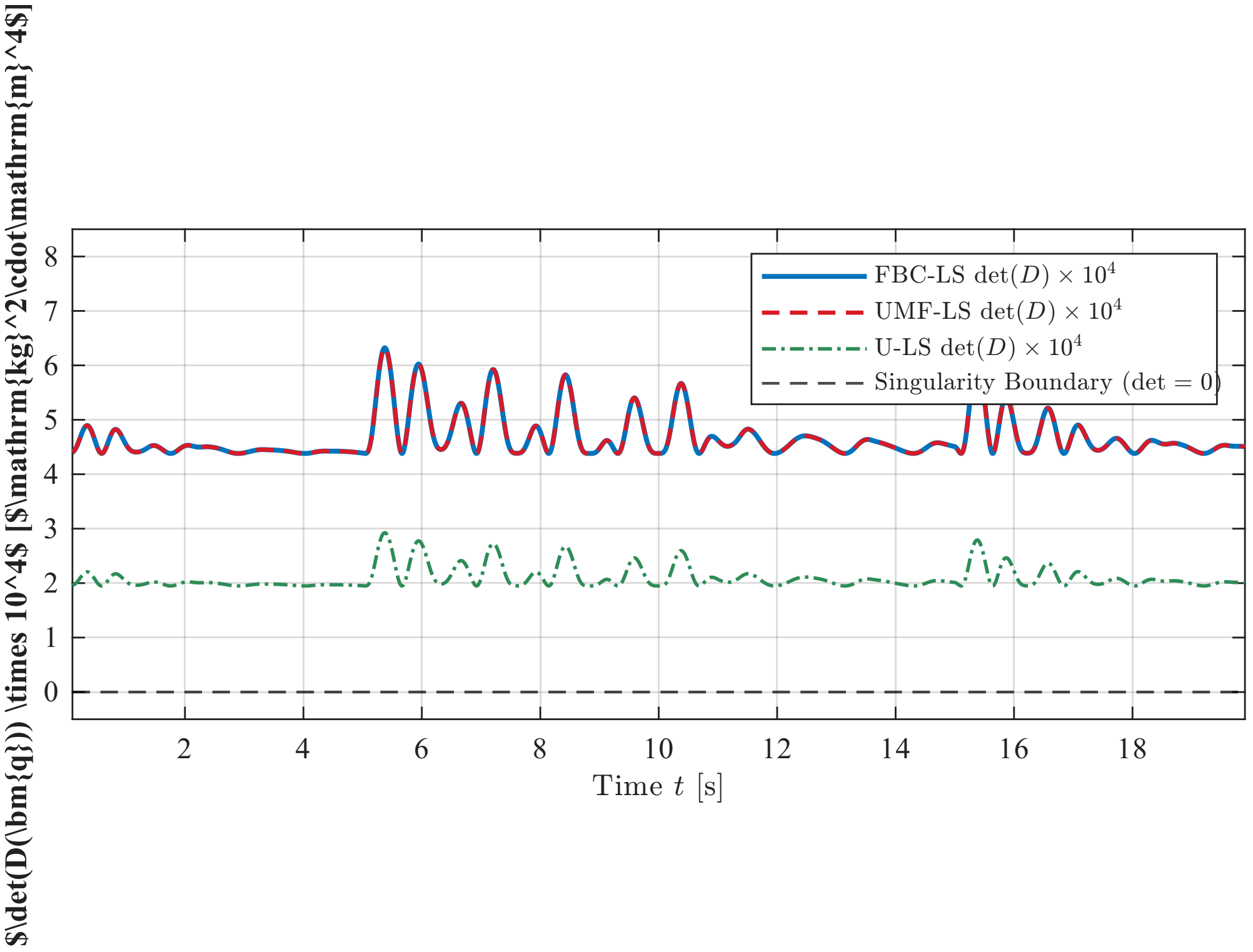

$\det(D(\bm{q})) \times 10^4$ [$\mathrm{kg}^2\cdot\mathrm{m}^4$]
8
7
6
5
4
3
2
1
0
2
4
6
8
10
12
14
16
18
Time t [s]
FBC-LS $\det(D) \times 10^4$
UMF-LS $\det(D) \times 10^4$
U-LS $\det(D) \times 10^4$
Singularity Boundary ($\det = 0$)